REVIEW FOLLOW-UP

VoltSentry — Answers to the Review Doubts

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What we changed since the last review, and how to answer each question

What we added

1 · Real grid oscillation

Every charger now switches between full power and zero together, once a second, so the values change on screen.

2 · No false ML alarms

Fixed the simulator's meter timing and retrained the model on full charging sessions: 25 minutes, zero false alarms.

3 · Defences tab

A new dashboard tab showing where our firewall sits and every safeguard, with live counts.

4 · Tested

184 automated tests pass. Every change was checked live on the dashboard.

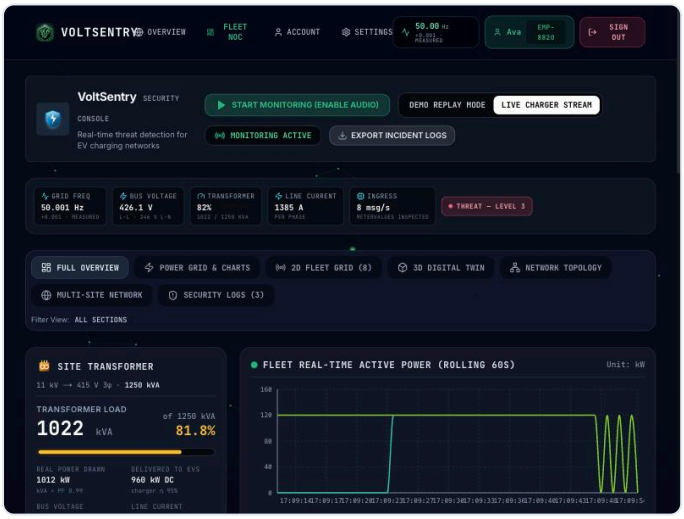
Doubt 1 — "In grid oscillation, the values don't change"

SAY

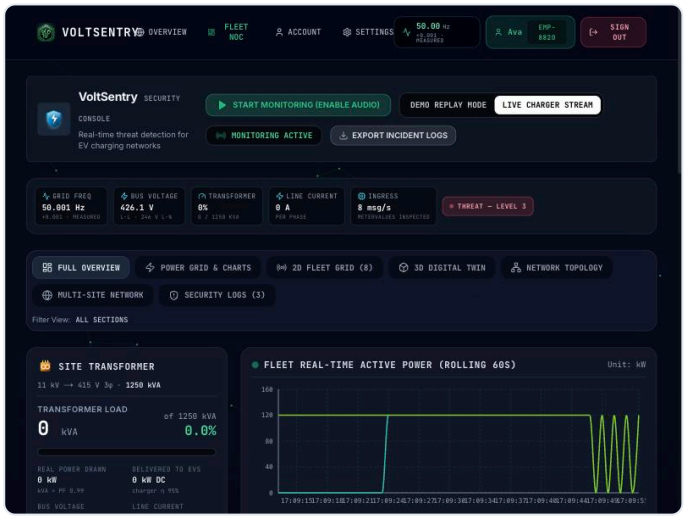
You were right, ma'am. Earlier the attack only sent start/stop messages without actually drawing power, so nothing moved on the screen. Now every charger switches between **full power and zero at the same moment, once a second** — like a real attack meant to stress the transformer. The transformer jumps between **82 % and 0 %**, the line current between **1385 A and 0 A**, and each charger card flips between **Charging 120 kW** and **Finishing 0 kW**. Our rule **R3** detects the pattern in about 1.5 seconds.

SHOW

Press `2` in the attack tool. Point at the **Site Transformer** panel and the charger cards — they change every second.



Second 1 — all chargers on: 1022 kVA (82 %), 1385 A



Second 2 — all chargers off: 0 kVA (0 %), 0 A

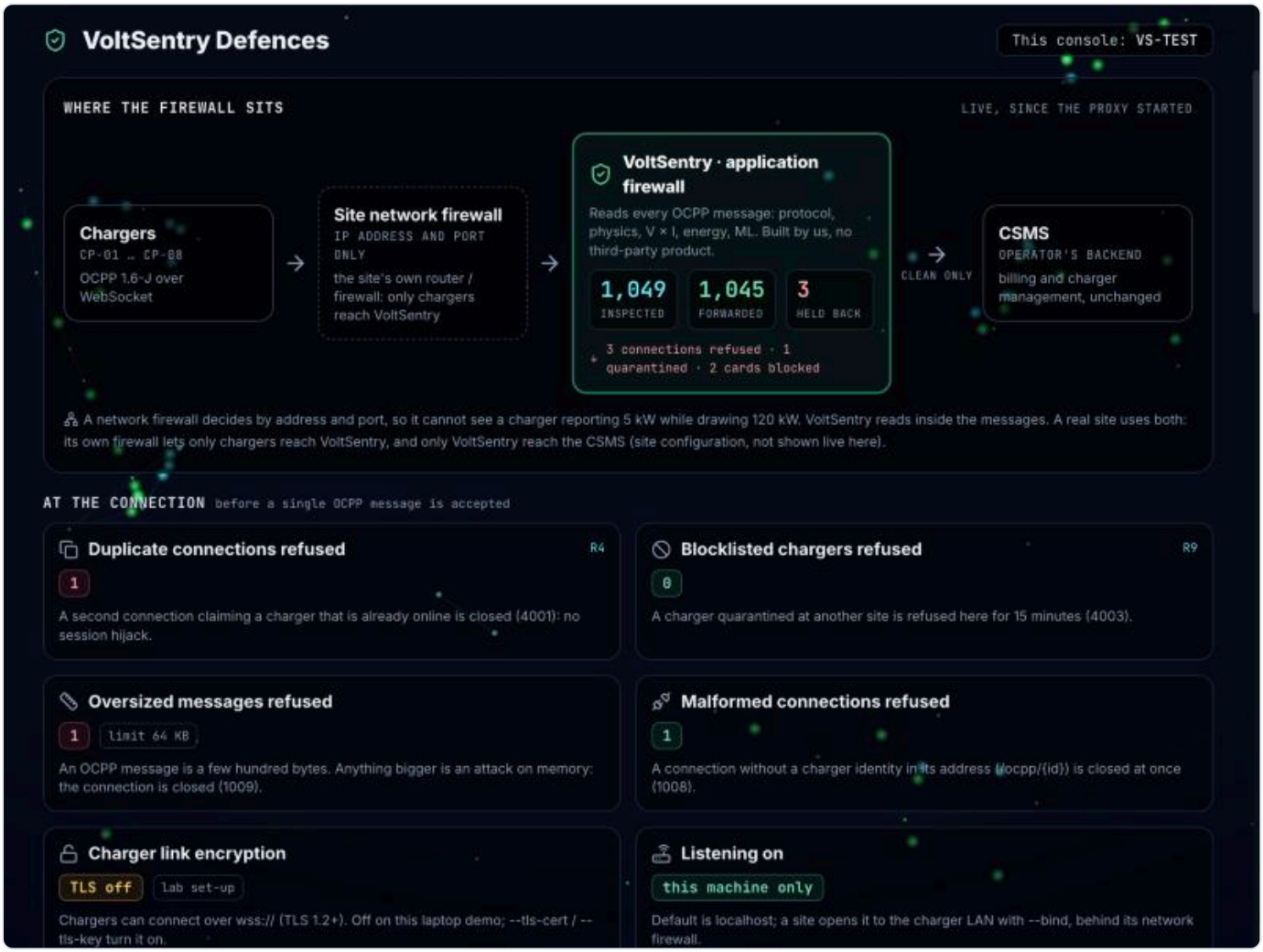
Doubt 2 — "What firewall are you using?"

SAY

VoltSentry itself is the firewall — an application-level firewall we built, which understands the OCPP charging protocol. A normal firewall only checks IP addresses and ports, so it cannot tell when a charger lies — for example reporting 5 kW while using 120 kW. VoltSentry reads every message and blocks the fake ones before they reach the server. A real site uses both: its normal network firewall in front, VoltSentry behind it.

SHOW

Open the **Defences** tab. The diagram at the top shows **Chargers** → **Network firewall** → **VoltSentry** → **Server**, with live counts of messages inspected, forwarded and blocked.



Defences tab after test attacks: 1,049 messages inspected, 1,045 forwarded, 3 held back; 3 connections refused, 1 quarantined, 2 cards blocked

Doubt 3 — "Apart from quarantine, what other safeguards do you use?"

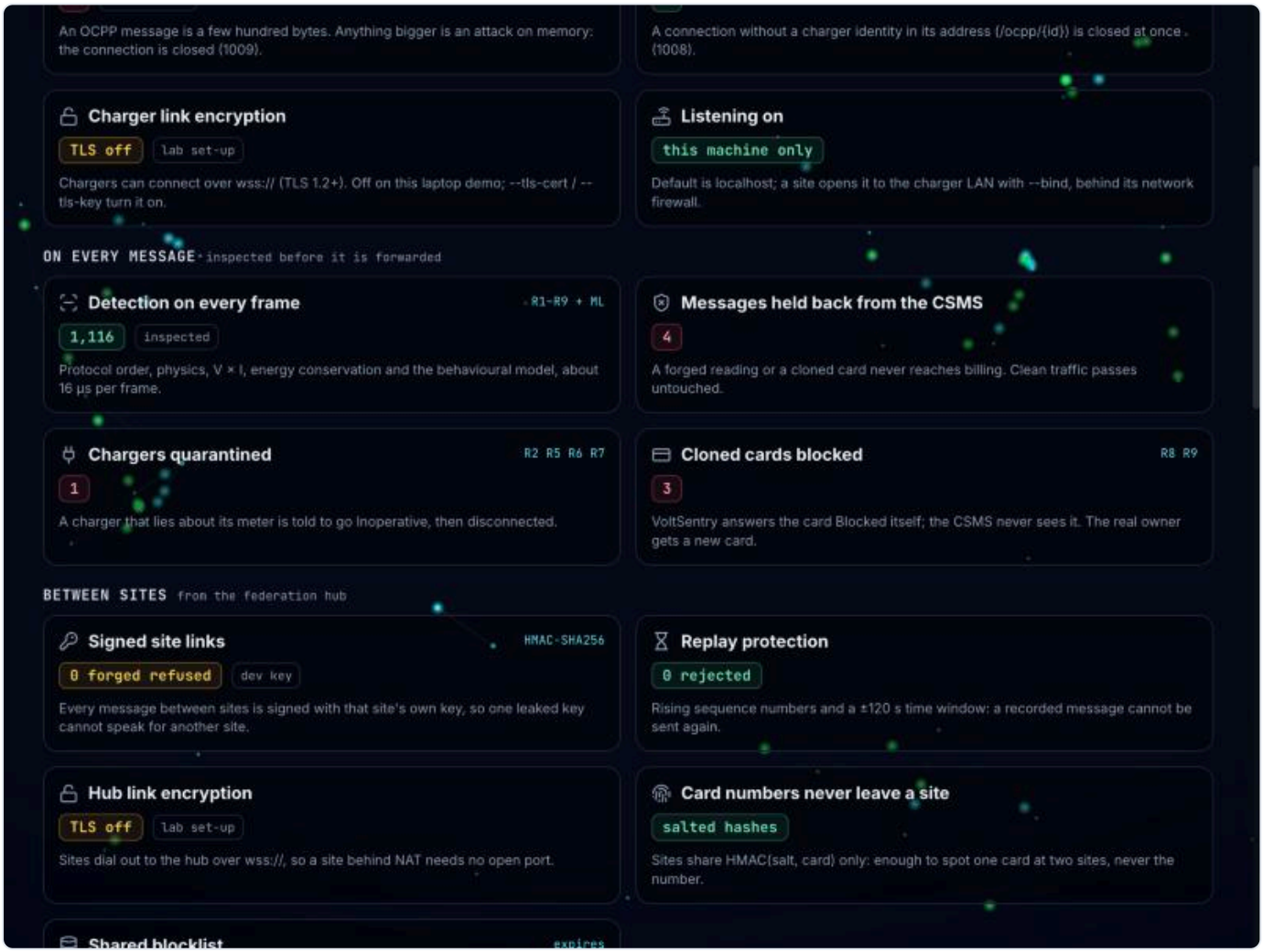
SAY

Several. We refuse fake duplicate charger connections, refuse chargers blocked by another site, refuse oversized or badly formed messages, and block cloned RFID cards. Messages between sites are signed and cannot be replayed, and card numbers are never shared — only scrambled hashes. New sites can start in monitor mode, and every attack is saved in a forensic log.

SHOW

On the **Defences** tab, scroll down. Each safeguard is a card with its live count.

WHERE	SAFEGUARD	WHAT IT STOPS
At the connection	Duplicate connection refused (R4) · blocklisted charger refused (R9) · messages over 64 KB refused · malformed connection refused · TLS encryption available	Session hijack, a known-bad charger, memory attacks, junk connections
On every message	9 rules + ML check every message · forged messages held back · cloned cards answered "Blocked"	Billing fraud, fake readings, card cloning
Between sites	Every message signed (HMAC-SHA256, each site's own key) · replay protection · card numbers shared only as hashes · blocklist that expires	Fake sites, replayed messages, leaking customer data
Operations	Monitor mode for new sites · forensic log of every incident · each site keeps working if the hub is down	Blocking real customers by mistake; losing evidence



Safeguard cards with live counts

If she asks what is missing — say it before she does: charger passwords and certificates (OCPP security profiles) and a real login for the dashboard are not built yet. They are marked **"planned"** on the same Defences screen. TLS works but is switched off on the laptop demo.

Before the review

1. Start the network: `python scripts/run_network.py`
2. Start the dashboard: `cd dashboard && npm run dev` → open `http://localhost:3000` → Live Charger Stream
3. Run a couple of attacks first (e.g. `python -m cli.attack -a 1`, then `-a r`) so the Defences counters are not zero
4. For doubt 1: `python -m cli.attack -a 2`, show the transformer, then `-a r`

VoltSentry · answers to the review doubts · every number above was measured on the running system.